

Final Report

Whitman Corlew

Introduction

Our project is an analysis of housing violations compared against residential property taxation in Montgomery County, Maryland. Our ultimate goal was to further the understanding of inequality and the tangible impacts it can have on people's lives. The result of our work on the data is a dashboard of relevant figures that will be described in full later. Our two datasets are both from the government website "Data Montgomery" which compiles civic data collections for easy public use. We selected "Real Property Tax 2024," a dataset consisting of around 345,000 cases and 75 columns. Each case represents an individual address associated with a residential property. Each column represents either some identification information about the property, or in most cases, it represents an individual type of tax, along with the amount of that tax, the property owner had to pay within the year of 2024. This can include standard things like state and federal taxes, along with taxes on things like water, or public services. This means each property might not necessarily pay for each type of tax. Fortunately, there is a column labeled "Total" which corresponds to the total amount of tax that property owners had to pay on that property in 2024. This is the column we will use for later analysis. This, along with a label of "Tax Class" which is a label used to identify and vary county tax based on where the property is located and what kind of district that location is. We will also use identifications like zip codes and address names. The other dataset we used is called "Housing Code Violations." This one includes over 935,000 individual cases and 23 columns. Each case represents a single violation on a property. The columns are much more varied than the previous set. They include information about the property(zip, state,address), along with information about the violation such as dates associated with the violation process, code references, and action required. We will be using most of these variables in our analysis. It also includes empty locational data in the form of latitude and longitude.

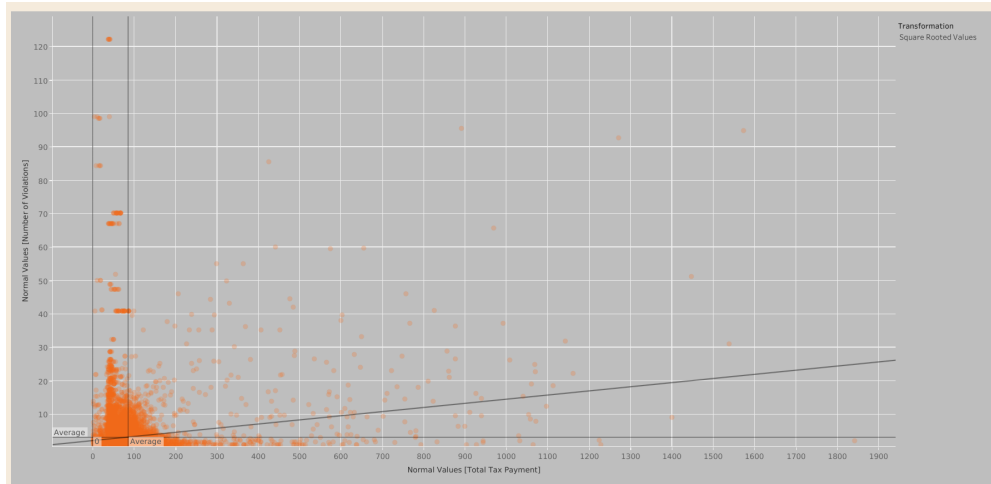
We used very few tools during the course of this project. The cleaning and wrangling were done entirely in Rstudio, due to our prior experience with the program. The visualization and analysis were done with Tableau. This is only possible because the project doesn't use any models. Finally, we used a tool published by the US Census Bureau, the Census Geocoder. This brings us into the cleaning summary section.

Cleaning

The bulk of the cleaning centered around the removal of confusing values from the data. The major issue with the tax dataset is that the numbers were entered such that the thousands were separated by commas, and each cell included a currency symbol (\$). This meant that when Rstudio was interpreting the data, it read these columns as text. After merging the two datasets together and removing the NA values from the total column, I used a selection loop to go through the columns and replace these nuisance values with null space. I also needed to be able to compare the data with the number of violations. Since there wasn't a discrete count in the data, I used a frequency table and merged it back onto the data. I then took only the unique addresses present in the data. Once that was done I needed to deal with the issue of having empty locational data. Without specific geo-locational points, I would be unable to plot this data onto a map. To solve this, I used the previously mentioned Geocoder. By splitting up this roughly 20,000 case tall dataset into 3 parts, I could use their location information(state, zipcode, city) in the geocoder to receive data about their precise geographic location in latitude and longitude. I then re-imported these back into the system and merged them to my small data. The rest of the data I would use in a separate dataframe for violation type.

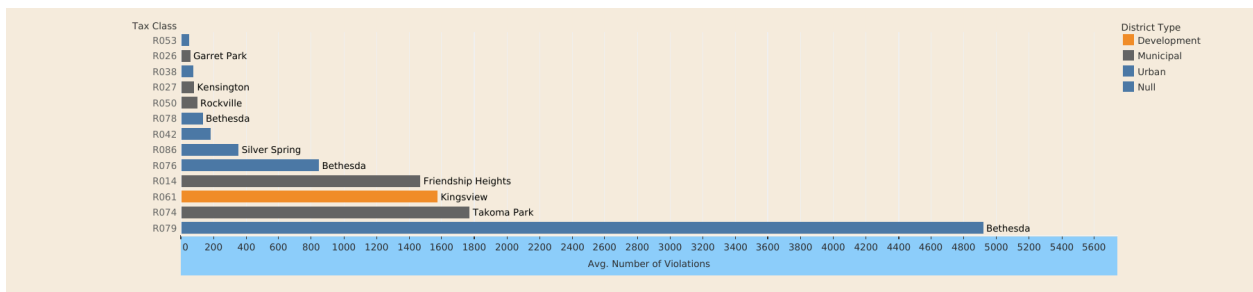
Final Product

The final result of the project is a dashboard. This dashboard is a collection of interactive visualizations designed to be explored yourself. The first visualization is a scatterplot comparing the total amount of tax paid for each property and the number of violations that property had.

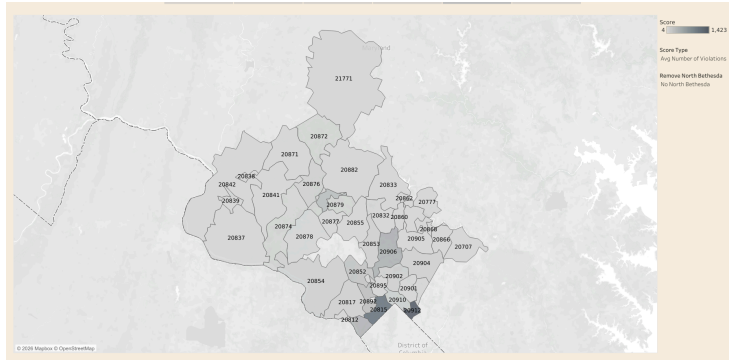


The original dashboard allows the user to choose which transformation of values to visualize. In this case the values are square rooted. This visualization indicates that there is some positive correlation between the amount paid and the number of violations.

The next visualization is a bar chart comparing the average number of violations the properties have between tax classes, with their colors determined by the type of district they are in.

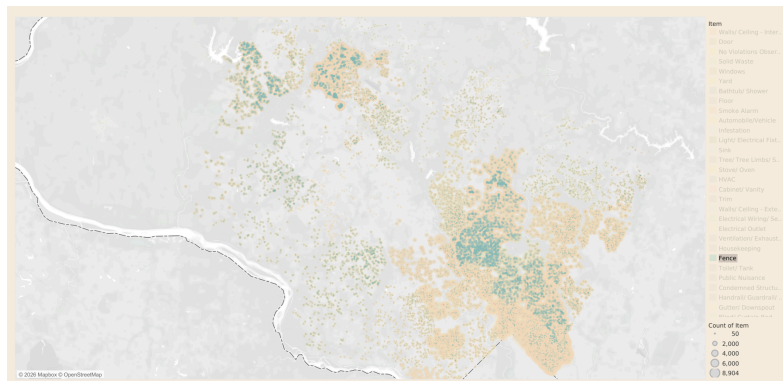


These are only the few highest offenders. They are made up of mostly Municipal and Urban district types. These are the types that would be found in cities and likely denote places with high density housing. The unlabeled 038 and 042 tax classes are large zones comprising multiple areas in MOCO and so do not have discrete names. They will be important in the next visualization. This next one is a dotplot indicating where each tax class can be found.



“Score” is used as a neutral term between different visualizations. North Bethesda has a huge average on account of having so few properties and skews the rest of the values. From this map we can see that the higher density areas like Rockville, Silver Spring, and Bethesda have the greatest average number of violations. These are also the most metropolitan areas.

The ultimate and hardest to transfer visualization is a dot plot of tens if not hundreds of thousands of points, each corresponding to an individual case. The addresses with more cases have larger points



This specific map is one showing where “Fence” type violations are the most common. They are shown in blue. They are seemingly most common in the more sub-urban regions between Bethesda and Rockville. There are many options for you to explore, each one revealing the socio-economic and infrastructural patterns across MOCO.

Reflection

Experience

This project was not particularly challenging. The most difficult parts were the numerous instances of problem solving, like needing to find locational data, and coming up with unique and clear visualization. The most rewarding part was the experience I gained in presentation. Even though I take issue with the concept of presenting a dashboard through a slideshow, I felt like I've learned a lot about how to structure a presentation and how to prepare yourself to speak. Given another opportunity, I would have chosen not to do a dashboard, as it makes for a terrible presentation.

Recommendation

A recommendation to the experts might be that your violation addressment efforts would be more efficient if they focused on the areas with the greater number of violations, cities. While not presented here, the dashboard shows where the amount of time it takes to close a case varies. A recommendation to the public is to explore the dashboard yourself. There is so much information that is impossible to fit into a document, with dozens of different kinds of violations and a number of hidden patterns.

Acknowledgements

I would like to thank my peers, my professors, and data Montgomery for all the help, education, and opportunity they have provided.